EXP NO:12

Determination and plotting of magnetic and electric field intensity due to infinite long conductor at any specific point located at H

```
Enter i value:3

Enter r value:0.006

  "Magnetic field intensity H = 79.577472"
Enter lambda value:3

Enter r value:0.004

  "Electric field intensity E = 1.348D+13"
```

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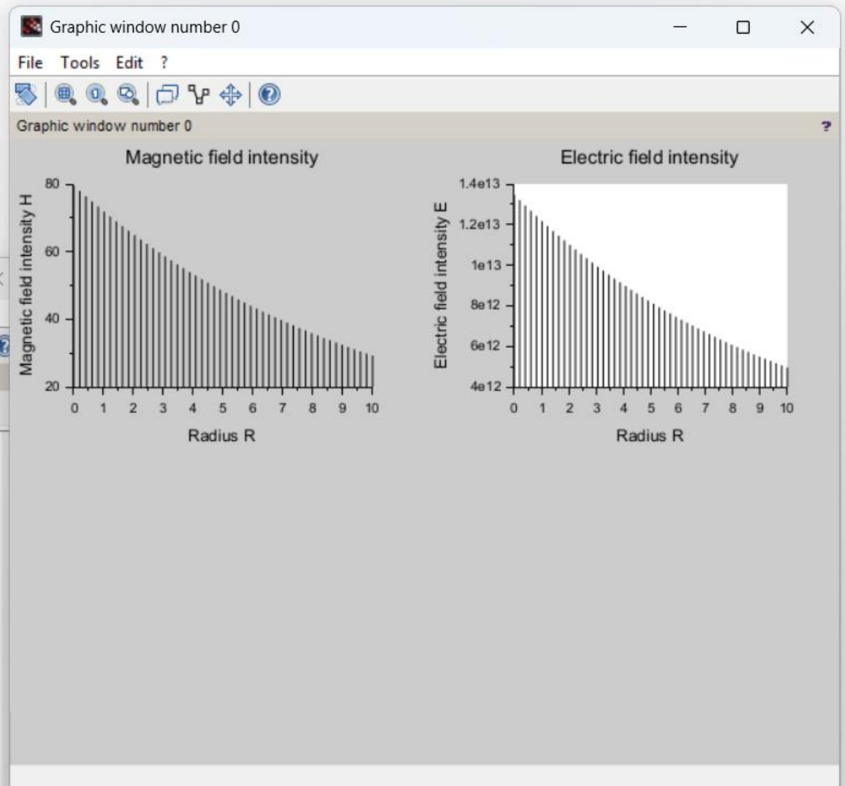
```
clc;
clear;
i = input("Enter i value:");
r = input("Enter r value:");
h = i / (2 * %pi * r);
disp("Magnetic field intensity H = " + string(h));
lambda = input("Enter lambda value:");
r = input("Enter r value:");   // reuse or enter new radius
Eo = 8.854e-12;
e = lambda / (2 * %pi * Eo * r);
disp("Electric field intensity E = " + string(e));
x = linspace(0,10,50);
y = h * exp(-0.1*x);
z = e * exp(-0.1*x);
figure();
subplot(2,2,1);
plot2d3(x,y);
title("Magnetic field intensity");
xlabel("Radius R");
ylabel("Magnetic field intensity H");
subplot(2,2,2);
plot2d3(x,z);
title("Electric field intensity");
xlabel("Radius R");
ylabel("Electric field intensity E");
```



Calculation:

Exp:- 12

$i = 3$ A

$r = 6$ mm $= 0.006$

$\lambda = 3$ m

$z = 4$ mm $= 0.004$

$\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

Magnetic field intensity (H):-

Formula:- $H = \dfrac{i}{2\pi r}$

Calculation:- $H = \dfrac{3}{2 \times 3.1416 \times 0.006} = \dfrac{3}{0.0377} = 79.6$ A/m

Electric field intensity E

Formula: $E = \dfrac{\lambda}{2\pi \varepsilon_0 r}$

$2\pi \varepsilon_0 r = 2 \times 3.1416 \times (8.854 \times 10^{-12}) \times 0.04 = 2.224 \times 10^{-13}$

$E = \dfrac{3}{2.224 \times 10^{-13}} = 1.35 \times 10^{13}$ V/m

CASE:1

To examine how the magnetic and electric field intensities change with varying distances from an infinitely long current-carrying conductor.

```
 "Distance  (m)  | Magnetic Field  (T)  | Electric Field  (N/C)"
  0.10        | 2.000000e-05    | 1.797548e+05
  1.00        | 2.000000e-06    | 1.797548e+04
  5.00        | 4.000000e-07    | 3.595097e+03
 10.00        | 2.000000e-07    | 1.797548e+03

-->
```

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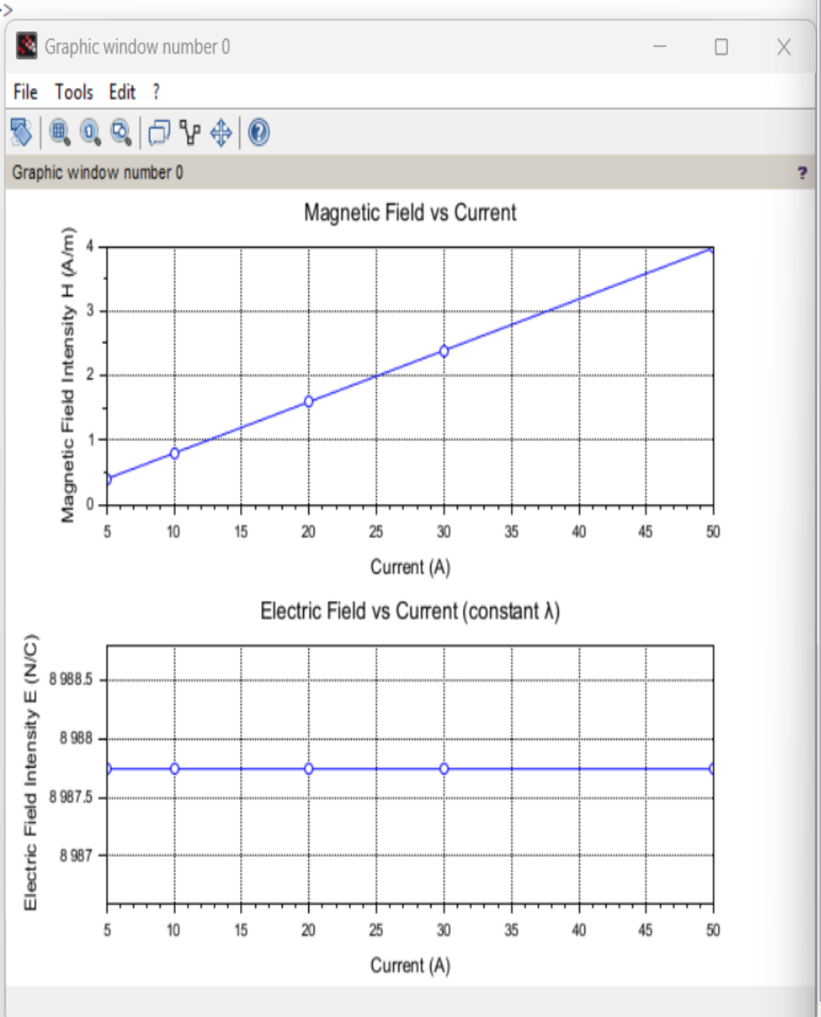
exp333.sci

```
clc;
clear;

// Constants
mu_0 = 4*%pi*1e-7;       // Permeability of free space (T·m/A)
epsilon_0 = 8.854e-12;   // Permittivity of free space (F/m)
I = 10;                  // Current (A)

// Distances from the conductor
distances = [0.1, 1, 5, 10]; // meters
num_distances = length(distances);

// Initialize arrays
B_field = zeros(1,num_distances);
E_field = zeros(1,num_distances);

// Assume a line charge density λ (C/m) for electric field
lambda = 1e-6; // example value

// Calculate fields
for i = 1:num_distances
    r = distances(i);
    B_field(i) = (mu_0*I)/(2*%pi*r);           // Magnetic field
    E_field(i) = lambda/(2*%pi*epsilon_0*r);   // Electric field
end

// Display results
disp("Distance (m) | Magnetic Field (T) | Electric Field (N/C)");
for i = 1:num_distances
    mprintf("%6.2f       | %12.6e   | %12.6e\n", distances(i), B_field(i), E_field(i));
end
```

CASE:2 To study how the magnetic and electric field intensities change for an infinitely long conductor at a fixed distance H but with varying currents.

```
"Current (A) | Magnetic Field Intensity (A/m) | Electric Field Intensity (N/C
   5     |        3.978874e-01       |        8.987742e+03
  10     |        7.957747e-01       |        8.987742e+03
  20     |        1.591549e+00       |        8.987742e+03
  30     |        2.387324e+00       |        8.987742e+03
  50     |        3.978874e+00       |        8.987742e+03

-->
```



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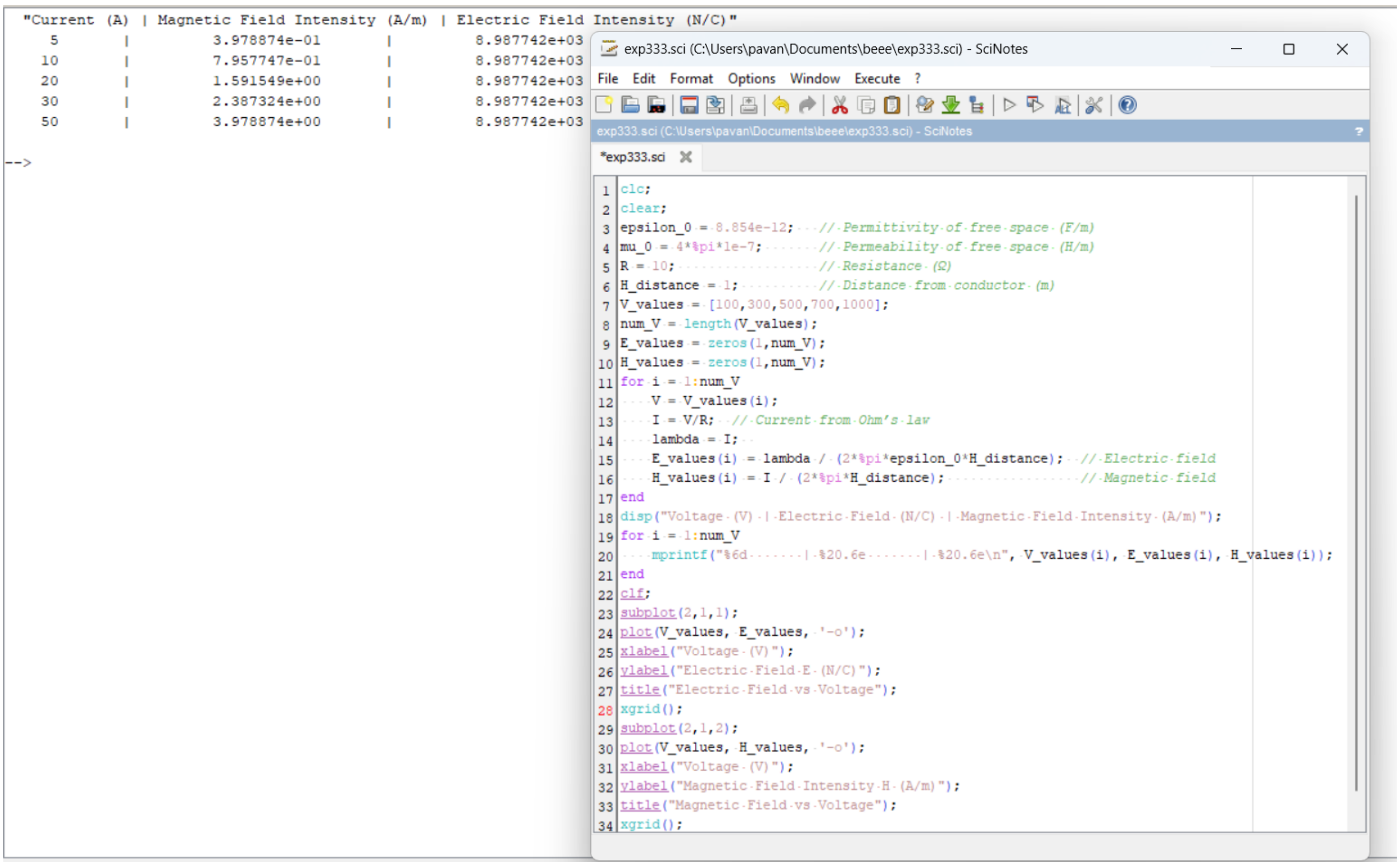
```
clc;
clear;
H = 2;                  // Distance from conductor (m)
epsilon_0 = 8.854e-12;   // Permittivity of free space (C^2/(N·m^2))
lambda = 1.0e-6;        // Line charge density (C/m)
currents = [5,10,20,30,50];
num_currents = length(currents);
magnetic_field_intensity = zeros(1,num_currents);
electric_field_intensity = zeros(1,num_currents);
for i = 1:num_currents
    I = currents(i);
    magnetic_field_intensity(i) = I/(2*%pi*H);                 // H field
    electric_field_intensity(i) = lambda/(2*%pi*epsilon_0*H);  // E field
end
disp("Current (A) | Magnetic Field Intensity (A/m) | Electric Field Intensity (N/C)");
for i = 1:num_currents
    mprintf("%6d       | %20.6e       | %20.6e\n", currents(i), magnetic_field_intensity(i), electric_field_intensity(i));
end
clf;
subplot(2,1,1);
plot(currents, magnetic_field_intensity, '-o');
xlabel("Current (A)");
ylabel("Magnetic Field Intensity H (A/m)");
title("Magnetic Field vs Current");
xgrid();
subplot(2,1,2);
plot(currents, electric_field_intensity, '-o');
xlabel("Current (A)");
ylabel("Electric Field Intensity E (N/C)");
title("Electric Field vs Current (constant λ)");
xgrid();
```

case 3: To analyze the magnetic and electric field intensities due to an infinitely long conductor when there is a potential difference (voltage) across the conductor, simulating a capacitive effect.



```
"Current (A) | Magnetic Field Intensity (A/m) | Electric Field Intensity (N/C)"
   5      |          3.978874e-01        |          8.987742e+03
  10      |          7.957747e-01        |          8.987742e+03
  20      |          1.591549e+00        |          8.987742e+03
  30      |          2.387324e+00        |          8.987742e+03
  50      |          3.978874e+00        |          8.987742e+03

-->
```

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```
clc;
clear;
epsilon_0 = 8.854e-12;   // Permittivity of free space (F/m)
mu_0 = 4*%pi*1e-7;       // Permeability of free space (H/m)
R = 10;                  // Resistance (Ω)
H_distance = 1;          // Distance from conductor (m)
V_values = [100,300,500,700,1000];
num_V = length(V_values);
E_values = zeros(1,num_V);
H_values = zeros(1,num_V);
for i = 1:num_V
    V = V_values(i);
    I = V/R;  // Current from Ohm's law
    lambda = I;
    E_values(i) = lambda / (2*%pi*epsilon_0*H_distance);  // Electric field
    H_values(i) = I / (2*%pi*H_distance);                 // Magnetic field
end
disp("Voltage (V) | Electric Field (N/C) | Magnetic Field Intensity (A/m)");
for i = 1:num_V
    mprintf("%6d       | %20.6e       | %20.6e\n", V_values(i), E_values(i), H_values(i));
end
clf;
subplot(2,1,1);
plot(V_values, E_values, '-o');
xlabel("Voltage (V)");
ylabel("Electric Field E (N/C)");
title("Electric Field vs Voltage");
xgrid();
subplot(2,1,2);
plot(V_values, H_values, '-o');
xlabel("Voltage (V)");
ylabel("Magnetic Field Intensity H (A/m)");
title("Magnetic Field vs Voltage");
xgrid();
```